KNPHI Laboratory Preparedness for Ebola Virus Disease (EVD)

Kenya

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Session Objectives

1 Review Current Preparedness

Assess county lab preparedness from EVD matrix (September 2025) and identify critical gaps

2 Clarify Laboratory Roles

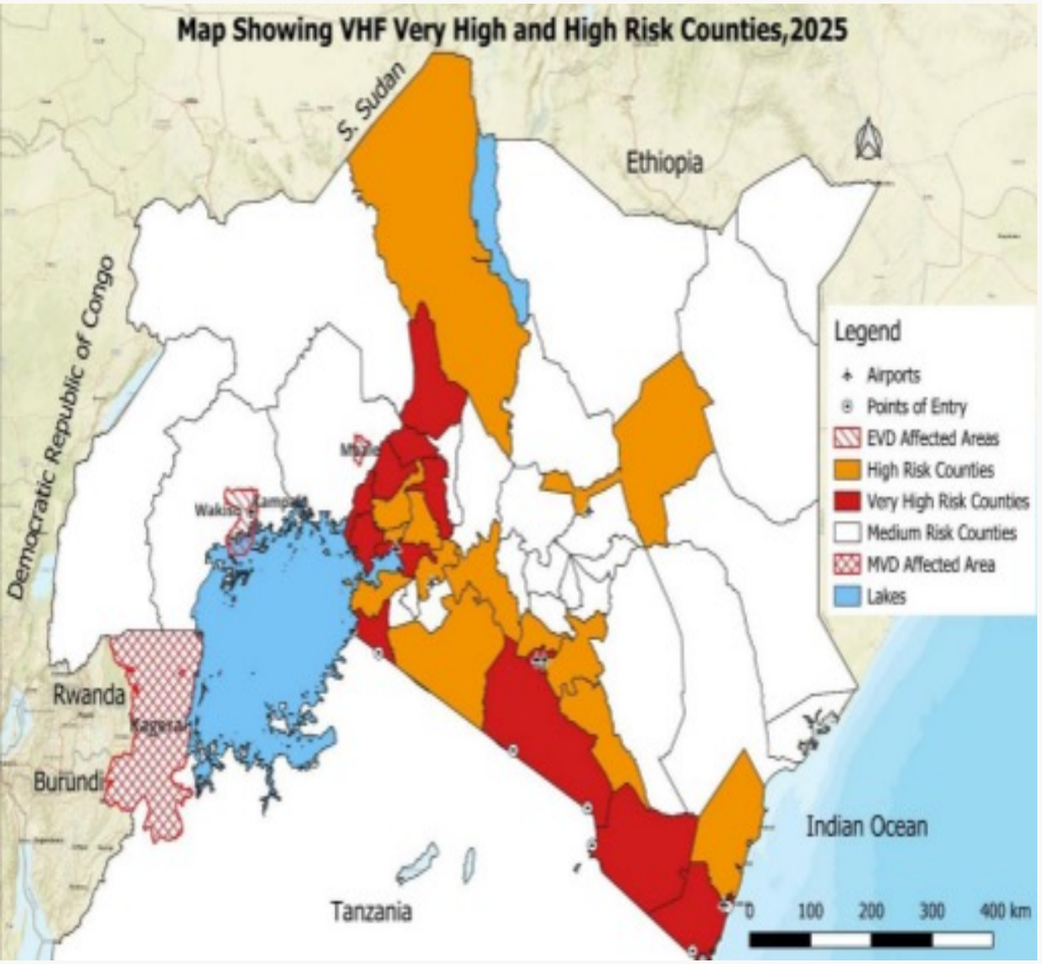
Define clear laboratory roles and establish robust referral pathways for EVD specimens

3 Agree Priority Actions

Set priority actions with specific timelines and assign responsible persons for implementation

4 Confirm Requirements

Validate SOPs, commodities, quality assurance, and biosafety requirements for EVD response



Situation Overview

National Context

13 high-prevalence VHF counties targeted for enhanced preparedness.
Lab coordination point at the county is through CMLC

Key Risk Drivers:

- **Active surveillance signals**
- **Cross-border movement patterns**
- **Previous VHF events in region**

Current Status

Matrix reviewed September 2025 reveals gaps in referral coverage, commodities, and quality assurance systems.

Immediate action required to address critical preparedness gaps before potential outbreak scenarios

Laboratory Preparedness Status

Current assessment reveals mixed readiness across core laboratory elements:

Testing Capacity

Limited EVD-validated assays at county level. Referral pathways established but require strengthening.

Specimen Referral

Incomplete referral mapping identified. Chain-of-custody procedures need standardization.

Biosafety & QA

PPE stockouts reported. Limited EQA participation across county laboratory network.

Element	Status	Priority	Action Needed
Workforce	🟡 Amber	High	Training
Commodities	🔴 Red	Critical	Immediate
Biosafety	🟡 Amber	High	SOPs

Laboratory Roles & Responsibilities

County Laboratories

- **Safe specimen collection and packaging**
- **Interim testing (if validated)**
- **Timely referral to reference labs**
- **Strict IPC adherence**

CMLC Leadership

- **Lead county ownership of operational plan**
- **Coordinate county multi-stakeholder response**
- **Ensure resource mobilization**

NPHL, KEMRI-DSLP, KEMRI-WRIAR, and KEMRI-VHF

- **Confirmatory testing services**
- **Quality assurance oversight**
- **Reagent provision and assay validation**
- **Technical mentorship and support**

Partners

- **Supply chain support**
- **Capacity building and training**
- **Logistics coordination**

Specimen Referral Network

Streamlined pathway ensures rapid, safe specimen transport from collection to confirmation:

Collection at Facility

Proper sample collection with safety protocols

Packaging & Chain-of-Custody

Triple packaging, documentation, cold chain maintenance

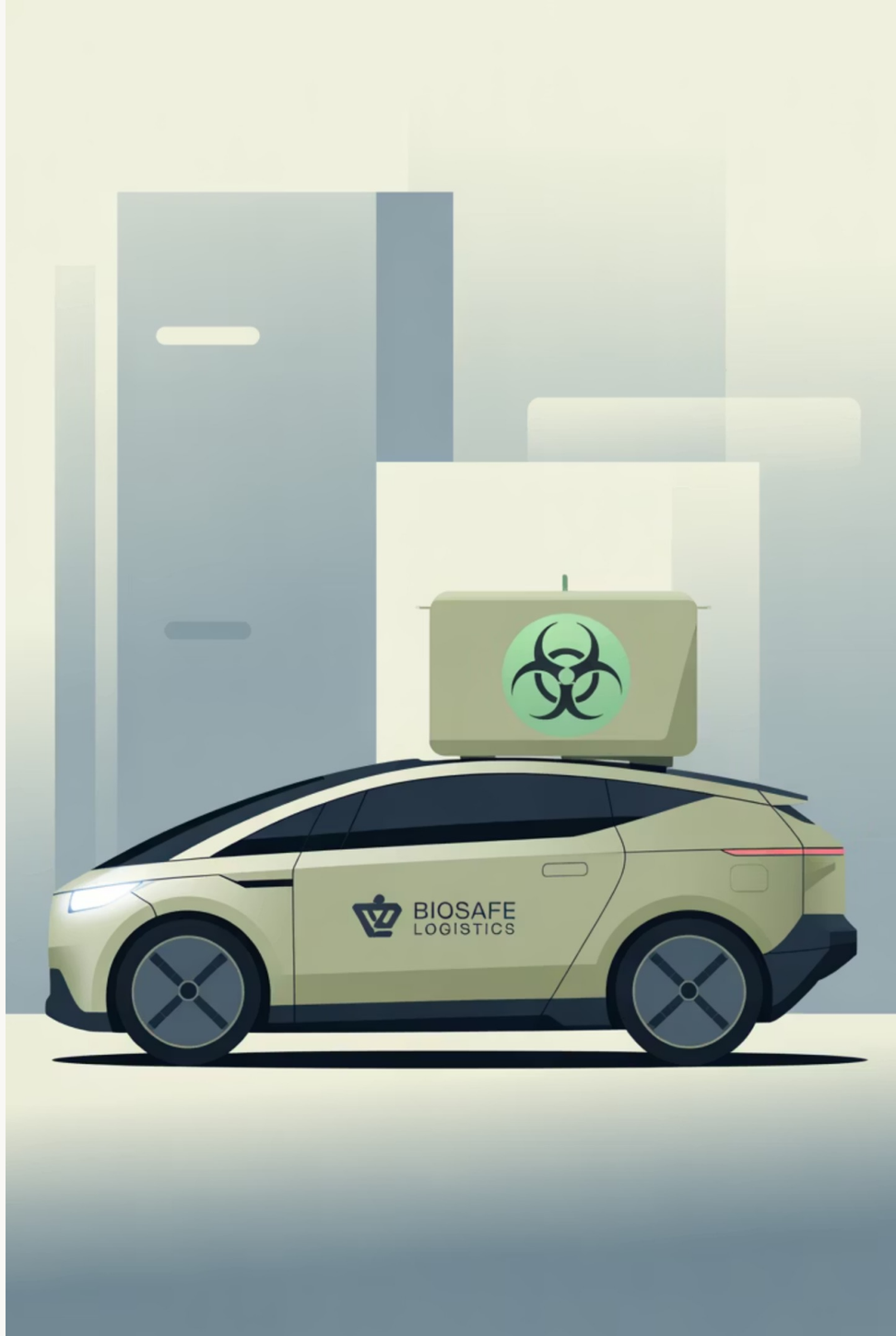
County Hub

Consolidation point for quality control and dispatch

Courier to NPHI Core Reference Laboratory(NPHL)

Secure transport to the reference laboratory

Timeliness Targets: Samples to referral hub within 6 hours; to NPHL within 24-48 hours



Sample Collection & Transport SOPs

Approved Sample Types

- **Whole blood (5-10ml EDTA tube)**
- **Serum (3-5ml clot tube)**
- **Oral fluid (if blood unavailable)**
- **Post-mortem skin biopsy**

Triple Packaging Requirements

1. **Primary container (leak-proof)**
2. **Secondary packaging with absorbent**
3. **Outer shipping container with labels**

Cold Chain & Documentation

Maintain 2-8°C throughout transport. Complete chain-of-custody forms with collector signature, time stamps, and recipient confirmation.

Emergency Contacts:

NPHL: +254-713269010,

Action Point: Confirm SOP availability in all facilities and distribute pocket guides

Testing Strategy & Diagnostic Capacity

EVD testing performed exclusively at designated reference laboratories with validated assays and trained personnel.

Screening Tests

Initial RT-PCR screening at designated reference labs only

Confirmatory Testing

Secondary assay confirmation using different target genes

Reagent Support

NPHI and US CDC provides validated reagents with stock monitoring systems

Assay Type	Required Reagents	Current Availability
RT-PCR Screening	**Primers, probes, controls**	🟡 **Limited stock**
Confirmatory RT-PCR	**Alternative targets**	🟢 **Available**
Antigen Detection	**Rapid test kits**	🔴 **Out of stock**

Fastrack the World Bank procurement and seek Donations from Africa CDC
Abdi: to share list of Lab commodities(RT-PCR)

Quality Assurance & Result Reporting

Internal QC

Daily controls with documented SOPs and performance monitoring

EQA/PT Participation

Quarterly proficiency testing enrollment for all testing sites



Performance Monitoring

Turnaround time targets and KPI dashboard tracking

Result Reporting

Lab → County Surveillance → NPHI-NPHL → WHO within defined timeframes

Action: Enroll four testing labaratories in EQA panels and establish reporting SLAs with a 24-hour result turnaround target. EQA/PT will be supported through interlaboratory comparison coordinated by the QA unit

Biosafety, Biosecurity & IPC

Minimum Biosafety Measures

- **BSL-2 minimum for sample handling**
- **Full PPE including face shields**
- **Restricted laboratory access**
- **Negative pressure ventilation preferred**

Waste Management

- **Autoclave all contaminated materials**
- **10% bleach decontamination**
- **Secure waste disposal pathway**

Biosecurity Requirements

- **Secure sample storage (-80°C freezers)**
- **Access control and inventory logs**
- **Chain-of-custody documentation**
- **Authorized personnel only**

Action: Biosafety focal person to schedule refresher training/TOT within 2 weeks to support ISRS training rollout



Workforce & Training Requirements



Required Competencies

Specimen handlers, lab technologists, and couriers must demonstrate proficiency in EVD sample protocols, biosafety measures, and emergency procedures.



Planned Training Program

Comprehensive reorientation on collection/packaging procedures, TOT sessions, and county-level emergency response drills scheduled for implementation.



Surge Capacity Plan

Rapid mobilization protocol with partner support, clear escalation pathways, and 24-hour response timeline for emergency situations.

Action: Confirm training dates, finalize participant lists, and designate CMLC as TOT lead coordinator. NPHL QA unit and Virology Lab Manager to perform competency assessment for National and county staff and document list of staff to support various Lab component

Commodities & Logistics

Priority Prepositioning Items

PPE Sets

Complete protective equipment including gowns, gloves, face shields, and respirators for 30-day supply

VHF Sample Kits

Triple packaging materials, collection tubes, transport media, and chain-of-custody forms

Testing Reagents

RT-PCR reagents, controls, and consumables with cold chain storage requirements

Cold Chain Equipment

Insulated transport boxes, temperature loggers, and backup cooling systems

Stock Monitoring: Weekly inventory checks with automated reorder triggers. Contact NPHL logistics coordinator for emergency resupply.

Action: A commodity logging and tracking system is to be implemented through inclusion into the LIMS. This is actioned to APHL

Risk Communication & Coordination

Laboratory's Communication Role

- **Provide timely, accurate result reporting**
- **Maintain clear communication channels**
- **Avoid creating public panic through measured messaging**
- **Support evidence-based decision making**

Key Coordination Channels

- **CMLC - County coordination lead**
- **County Disease Surveillance Unit**
- **NPHL - Technical oversight**
- **Partner focal points - Resource support**

Emergency Contact List

Contact	Role	Phone/Email
Dr. Kingwara Leonard	**NPHL Lead**	**leonard.kingwara2gmail.com**
Dr Naomi Lucchi	**CDC Partner**	**nlucchi@cdc.gov**
David Mungai	**FHI-360**	DNdungu@fhi360.org

Action Plan & Timelines

Immediate 2-week sprint to address critical preparedness gaps:

1

Days 1–3: Mapping & SOPs

Action: Map county referral routes and distribute EVD sample SOPsResponsible: CMLC with NPHL supportDeliverable: Complete referral pathway documentation

2

Days 2–5: Commodity Prepositioning

Action: Preposition PPE and VHF sample kits Responsible: Logistics focal with partner supportDeliverable: 30-day supply inventory confirmed

3

Days 5–10: Training & Capacity

Action: Conduct TOT and biosafety refresher trainingResponsible: CMLC training coordinatorDeliverable: Certified personnel roster

4

Days 10–14: QA & Monitoring

Action: Enroll in EQA programs and establish monitoringResponsible: Lab quality officerDeliverable: KPI dashboard operational

Monitoring & Key Performance Indicators

95%	24h	100%	30
Proper Packaging Target	Sample TAT to NPHL	EQA Participation	PPE Stock Days
Percentage of samples meeting triple packaging requirements with complete documentation	**Average turnaround time from county collection to NPHL receipt**	**All designated labs enrolled in quarterly proficiency testing programs**	**Minimum days of PPE stock maintained at county level for surge response**

Reporting Cadence: Weekly summary reports to NPHL and county emergency operations center with trend analysis and corrective actions.

Action: Nominate dedicated monitoring officer and finalize reporting template within 3 days.

Challenges & Mitigation Strategies

Reagent Supply Constraints

Challenge: Limited reagent availability and supply chain disruptions
Mitigation: NPHI, FHI and CDC reagent backfill support and alternative supplier identification
Support Needed: Emergency procurement authorization

Transport Funding Gaps

Challenge: Insufficient funding for specimen courier services
Mitigation: Pooled sample routing and alternate courier arrangements
Support Needed: Emergency transport fund allocation

Workforce Capacity

Challenge: Limited trained personnel for surge response
Mitigation: Rapid competency checks and cross-training programs
Support Needed: Additional training resources and mentorship



County Readiness Checklist

One-page verification tool for immediate preparedness assessment:

- [ ] ✓ SOPs Available

  EVD sample collection, packaging, transport, biosafety, and waste management SOPs distributed to all facilities

- [ ] ✓ Chain-of-Custody Forms

  Standardized forms available at all collection points with trained personnel for completion

- [ ] ✓ PPE Stocked (30 Days)

  Complete PPE sets for specimen collection and handling with inventory tracking system

- [ ] ✓ Referral Hub & Courier

  Designated county referral hub with confirmed courier contact and transport arrangements

- [ ] ✓ EQA Enrollment

  All testing laboratories enrolled in external quality assessment programs

- [ ] ✓ Trained Focal Persons

  Named and trained focal persons for biosafety, quality assurance, and logistics coordination

Print as pocket checklist for CMLC and laboratory leads - verify all items before declaring readiness

Reference Materials & Annexes

Comprehensive documentation package to be shared post-meeting:



EVD Preparedness Matrix

Complete September 2025 matrix with annotations, gap analysis, and recommended actions for county implementation



Standard Operating Procedures

Sample collection, packaging, transport, biosafety protocols, and waste management procedures with step-by-step guidance



Forms & Templates

Chain-of-custody forms, sample submission templates, commodity order forms, and reporting templates



Contact Directory

Complete contact list including NPHL, CDC partners, emergency coordinators with phone numbers and email addresses



Assessment Documents

Detailed assessment reports referenced in matrix with recommendations and implementation guidance

Next Steps & Coordination

Immediate Actions

Confirm next coordination meeting: [Date/Time TBD]

- **Establish weekly check-in calls during 2-week sprint**
- **Activate emergency communication channels**
- **Begin implementation of agreed action plan**

Point of Contact for Issues

Primary: Specimen Referral CoordinatorPhone: +254-732353535 Email: manager.eockenya@gmail.com

Backup: NPHL Emergency LinePhone: +254-713-269-010 (24/7)

Result Email: Email sharing results with facilities: xxxxxxxx@nphi.co.ke

Remember: This is a critical preparedness initiative. Every action taken today strengthens our collective ability to respond effectively to potential EVD threats.



Thank You